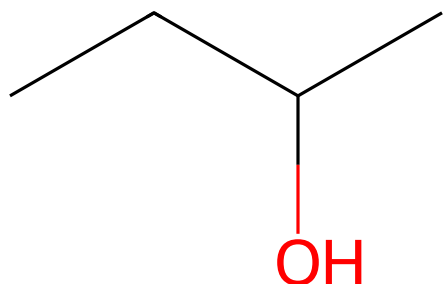


Organic chemistry practice: answer key

Skeletal structures: every corner and line end is a carbon, hydrogens on carbon are left out, and other atoms (O, Br) are written as letters. The grey code under each answer is its SMILES, which you can paste into the test page to compare with your drawing.

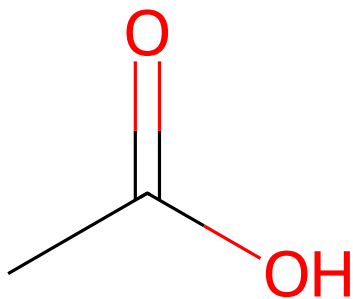


1. 2-butanol

A four-carbon zigzag: three lines. Every corner and every line end is a carbon, and hydrogens on carbon are not drawn.

Put one more line on the second carbon, up or down, and write OH at the end of it.

CCC(C)O

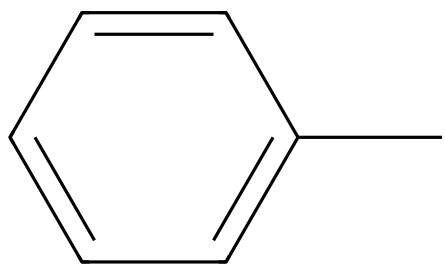


2. acetic acid

Two carbons: one line. The right-hand carbon gets a double line up to an O (that is C=O) and a single line out to OH.

C=O plus OH on the same carbon is a carboxylic acid.

CC(=O)O

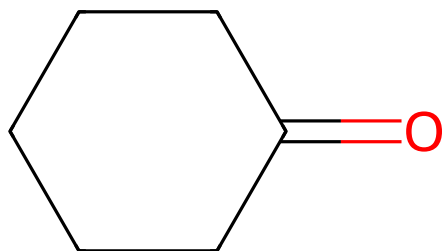


3. toluene

A hexagon with three alternating double lines inside it is a benzene ring.

One plain line off any corner is the CH₃ group. You do not write CH₃; the bare line end is the carbon.

Cc1ccccc1



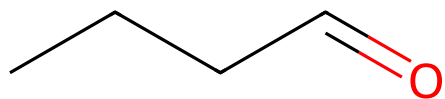
4. cyclohexanone

PCC turns a C–OH into a C=O. The ring stays a plain hexagon, with no double lines inside.

Draw a double line from one corner out to an O. The H of the OH is gone.

O=C1CCCCC1

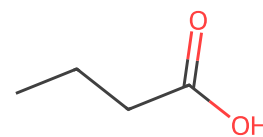
5. butanal



Start from the four-carbon zigzag of 1-butanol. PCC stops at the aldehyde, so the end carbon becomes C=O.

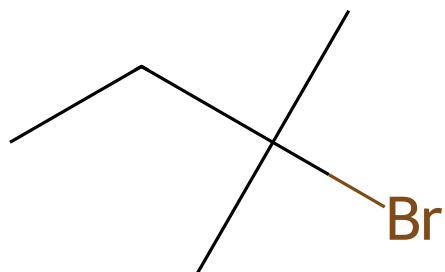
Common mistake: drawing butanoic acid (C=O and OH). That needs a stronger oxidant such as chromic acid.

CCCC=O



Not this: butanoic acid (over-oxidised)

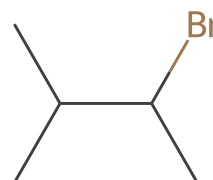
6. 2-bromo-2-methylbutane



Markovnikov: H goes to the alkene carbon that already has more H, so Br lands on the more substituted carbon, the one carrying the methyl.

Four-carbon zigzag; the second carbon gets two extra lines, one ending in nothing (the methyl) and one ending in Br. Common mistake: Br on the third carbon.

CCC(C)(C)Br



Not this: 2-bromo-3-methylbutane (anti-Markovnikov)